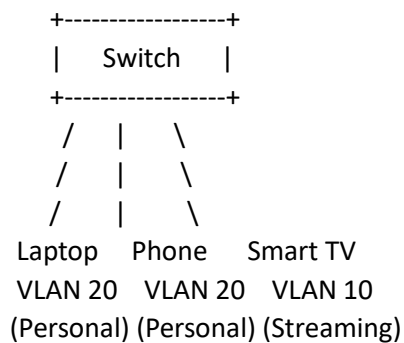


N+ - Network Fundamentals and building network

SWITCHING & VLAN

[Q-1] Draw a simple network diagram showing three devices (like your laptop, a friend's phone, and a smart TV) connected to a switch, and label which device is in which VLAN (e.g., VLAN 10 for streaming, VLAN 20 for personal devices).



VLAN Assignment

VLAN ID	Name	Devices
VLAN 10	STREAMING	Smart TV
VLAN 20	PERSONAL	Laptop, Phone

[Q-2] Configure a simulated switch using Cisco Packet Tracer or NetSim: create two VLANs (VLAN 10 and VLAN 20), assign three PCs to VLAN 10 and two PCs to VLAN 20, and verify that PCs in different VLANs cannot ping each other.
Hint: Use the 'vlan' and 'switchport access vlan' commands in the simulator CLI.

```
enable
configure terminal
```

```
vlan 10
name STREAMING
```

```
vlan 20
name PERSONAL
```

```
interface range fa0/1-3
```

```
switchport mode access  
switchport access vlan 10
```

```
interface range fa0/4-5  
switchport mode access  
switchport access vlan 20
```

```
end
```

PC Assignment

PC	Port	VLAN
----	------	------

PC1	Fa0/1	10
-----	-------	----

PC2	Fa0/2	10
-----	-------	----

PC3	Fa0/3	10
-----	-------	----

PC4	Fa0/4	20
-----	-------	----

PC5	Fa0/5	20
-----	-------	----

Verification

```
show vlan brief
```

Ping Test Results

Source	Destination	Result
--------	-------------	--------

PC1	PC2	Success
-----	-----	---------

PC1	PC4	Fail
-----	-----	------

PC2	PC5	Fail
-----	-----	------

[Q-3] In any network simulator, set up a scenario where you move a PC from VLAN 10 to VLAN 20. Show the before and after configuration commands and test connectivity to confirm the change.

Before

```
interface fa0/3  
switchport mode access  
switchport access vlan 10
```

PC3 belongs to VLAN 10.

After

```
interface fa0/3  
switchport mode access  
switchport access vlan 20
```

PC3 now belongs to VLAN 20.

Connectivity Test

Before Moving PC3

Source Destination Result

PC3	PC1	Success
-----	-----	---------

PC3	PC4	Fail
-----	-----	------

After Moving PC3

Source Destination Result

PC3	PC1	Fail
-----	-----	------

PC3	PC4	Success
-----	-----	---------

Explanation: After moving PC3 to VLAN 20, it can communicate with other devices in VLAN 20 but not with devices remaining in VLAN 10.

[Q-4] Write a short script or command list that would automate assigning 10 ports on a switch to VLAN 30, using the syntax of your chosen simulator (Packet Tracer, NetSim, or GNS3).
Constraint: Use a loop or range command if supported by the tool to avoid repeating the same command 10 times.

Configuration

```
enable
```

```
configure terminal
```

```
vlan 30
```

```
name VLAN30
```

```
interface range fa0/6-15
```

```
switchport mode access
```

```
switchport access vlan 30
```

Explanation

The interface range command allows multiple interfaces to be configured simultaneously. Instead of configuring each port individually, ports **Fa0/6 through Fa0/15** are assigned to VLAN 30 with a single command block.